

Measurement-shifting in the syntax: A rational solution to an agreement-interpretation puzzle?

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1 Introduction

In this work, I propose an amendment to previously-established architecture for measurement in natural language. While this proposal may be a radical departure from some long-held assumptions about measurement and semantics, I argue it out-competes alternatives for what is at present a seldom explored phenomenon. Empirically, the spotlight falls on FRACTION PARTITIVES (FPs), a nominal construction exemplified in (1). FPs which embed a plural DP are canonically understood to denote a subset of their complement whose cardinality is that of the complement multiplied by the fraction. Ergo, (1) could mean two out of six apples, three out of nine apples, etc. (see, e.g., [Ionin et al. 2006](#); [Wagiel 2021](#); [Benbaji-Elhadad and Wehbe 2024](#)).

(1) A third of the apples

For languages like American English, this understanding is fairly suitable. However, some languages exhibit an additional reading for FPs which has received very little attention in the literature. To illustrate, imagine a dingy studio apartment in Rome, Italy. Aerially, this apartment is an unequilateral hexagon, with two four-meter-wide walls and four one-meter-wide walls. Assuming the walls are of uniform height, the combined surface area of the apartment’s walls is $12 \times h$ square meters (where h is the apartment’s height). This is diagrammed in *fig. 1*.

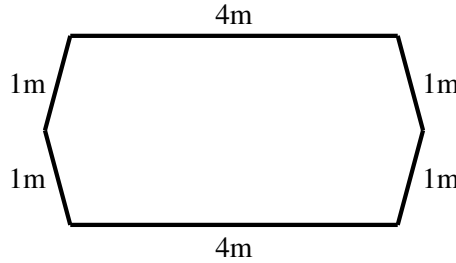


Figure 1: Floor plan of the studio apartment (not to scale)

Let’s say this apartment has developed a mold problem—blight has completely covered one third of the walls. If the Italian-speaking tenant of this unit called their landlord and said (2), this would be compatible with two scenarios. First, it may be that two out of the six walls are completely covered in mold (*fig. 2a*); this interpretation, which I presented above, is what I call the COUNT reading. Alternatively, it could be that one of the large walls (whose surface area is a third of that of the combined walls) is completely covered in mold (*fig. 2b*); I dub this interpretation, which is unavailable in American English, the MEASURE reading.¹

(2) la muffa ha coperto un terz-o del-le paret-i
 DET.F.SG mold has.3SG covered.3SG one third-M.SG of-DET.F.PL wall-F.PL
 ‘The mold has covered a third of the walls’ [Italian]

What is even more interesting is that Italian grammar has an inherent way to circumvent this confusion. If a FP is in a position of agreement, agreeing elements can copy the features of either the fraction (in this case, *terzo*

¹My use of *unambiguously* in *fig. 2a, 2b* should be clear; if the apartment layout was s.t. every member of the wall plurality was congruent (same size and shape), then there would be substantial overlap between the count and measure reading.

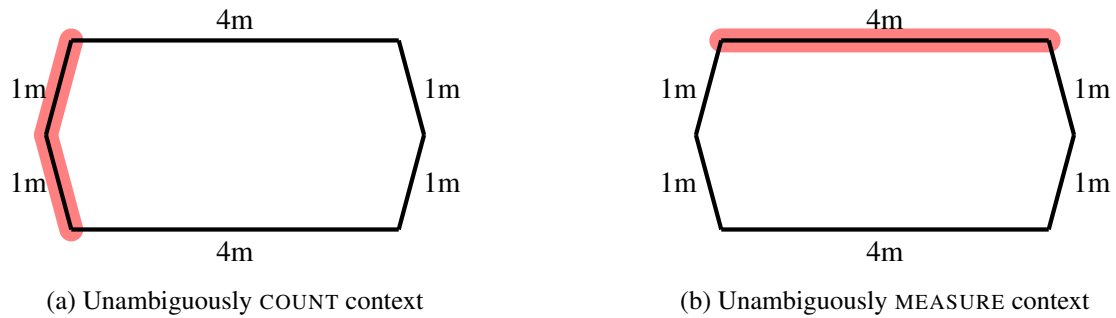


Figure 2: Two possible scenarios for mold coverage

(masculine singular)) or the complement of the FP (*le pareti* (feminine plural))). If the denominator is agreed with, only the measure reading is available. If instead agreement is with the complement, only the count reading is available.

- (3) ‘One third of the walls are covered by mold’ [Italian]
- a. (esattamente) un **terz-o** del-le paret-i è coperto di muffa
 (exactly) one third-M.SG of-DET.F.PL wall-F.PL be.3SG covered.M.SG by mold
 ✓M, ×C
- b. (esattamente) un **terz-o** del-**le** **paret-i** sono coperte di muffa
 (exactly) one third-M.SG of-DET.F.PL wall-F.PL be.3PL covered.F.PL by mold
 ×M, ✓C

Of course, the measure interpretation is compatible with other scenarios as well; the only constraint is that the cumulative square meterage of the moldy region be $4h\ m^2$. It could be that the lower third of each wall is covered in mold; In an even more dire accommodation situation where there are only two walls, it could be that two thirds of one wall are covered in mold. I use the dichotomy in *fig. 2a, 2b* for expository purposes.

This general pattern extends to other kinds of count nouns besides just *walls*. The same can be observed with more-transparently discrete pluralities like cakes, and even pluralities of abstract objects like films.

- (4) ‘A third of the cakes were eaten’ [Italian]
- a. **un** **terz-o** delle tort-e è stat-o mangiat-o
 a third-M.SG of.the.F.PL cake-F.PL be.3SG been-M.SG eaten-M.SG
 ✓M, ×C
- b. un **terz-o** **delle** **tort-e** sono stat-e mangiat-e
 a third-M.SG of.the.F.PL cake-F.PL be.3PL been-F.PL eaten-F.PL
 ×M, ✓C
- (5) ‘A third of the films were watched by John’ [Italian]
- a. **un** **terzo** de-i film è stato visto
 one third of-M.PL film be.3SG been.M.SG watched.M.SG
 ✓M, ×C
- b. un **terzo** de-**i** **film** sono stati visti
 one third of-M.PL film be.3SG M.PL watched.M.PL
 ×M, ✓C

Furthermore, this is not a phenomenon unique to Italian; it can be observed to some extent in many Indo-European languages. French and Greek exhibit exactly the Italian pattern.

- (6) ‘One third of the walls are covered by mold’ [French]
- a. **un** **tiers** des murs est couvert de moisissures
 one third of.DET.M.PL wall.M.PL COP.3SG covered.M.SG by mold
 ✓M, ×C

- b. un tiers **des** murs sont couverts de moisissures
 one third of.DET.M.PL wall.M.PL COP.3PL covered.M.PL by mold
 ×M, ✓C

(7) ‘One third of the pies were eaten’ [Greek]

- a. **ena trit-o** ton pit-on fagoθik-e
 one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3SG.PST
 ✓M, ×C
- b. ena trit-o **ton** **pit-on** fagoθik-an
 one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3PL.PST
 ×M, ✓C

A similar pattern can be observed with the fraction partitives of Russian and German, albeit with the caveat that a count reading is still available when Agreement is with the denominator.

(8) ‘One third of these² cakes were eaten’ [Russian]

- a. **(odna) tret’** étix pirog-ov byl-a s”eden-a
 one third.F.SG this.GEN.PL cake-GEN.PL be-F.SG eaten-F.SG
 ✓M, ✓C
- b. (odna) tret’ **étix** **pirog-ov** byl-i s”eden-i
 one third.F.SG this.GEN.PL cake-GEN.PL be-PL eaten-PL
 ×M, ✓C

(9) ‘One third of the cakes were eaten’ [German]

- a. **Ein Drittel** der Kuchen wurde aufgegessen
 one third DET.GEN.PL cake.GEN.PL be.PST.3SG eaten
 ✓M, ✓C
- b. Ein Drittel **der** **Kuchen** wurden aufgegessen
 one third DET.GEN.PL cake.GEN.PL be.PST.3PL eaten
 ×M, ✓C

Compare all of these cases again with American English—a language where only the count reading is available. Interestingly, when the complement of a fraction partitive is plural, agreement must necessarily be with that complement.

- (10) One third of the walls {*is/are} moldy
 ×M, ✓C

To recap, the generalization can be described as the following. When a DP-external element (e.g., a verb) agrees with the complement of a FP, the interpretation is necessarily a count one. When said element agrees with the denominator (in languages where circumstances allow agreement with a denominator), there is at least a measure interpretation. Whether or not there is also a count interpretation under fraction agreement is a point of cross-linguistic variation.³

The primary objective of the present study is to provide an account of this alternation specifically for Italian. The ideal proposal should be designed to secondarily explain why this alternation exists in some languages but not others, so I aim to do so. Ultimately, I propose a solution requiring an overhaul of how measurement is defined at LF; specifically, it is suggested that intensional operators can affect how variable measure functions are determined, and I further posit that these operators can enter the derivation in Narrow Syntax. This is a stark departure from previous work on the semantics of measure, which often assumes that the process is deterministic for pluralities and otherwise purely contextual.

²Consultant found the measure reading easier to attain with a demonstrative heading the complement. The measure reading is still marginally available without a demonstrative.

³The very fact that agreement is possible with the complement is mysterious. Much prior work on FPs provide a treatment where the fraction embeds the complement, yielding a DP-within-DP configuration. If we subscribe to a theory of Agree where locality resolves competition between multiple (bundles of) features (Chomsky, 1995), only agreement with the fraction should be possible. As is clear by this point, though, this is not what we see.

The paper is structured as follows. I establish background knowledge of the semantics of parthood and measurement in service of demonstrating how previous authors have derived the count reading in §2. In §3, I outline my own proposal given previous literature. §4 considers possible alternatives to my proposal; strengths and critical weaknesses are highlighted and contrasted with those of my own. §5 concludes.

2 Background: Deriving Count

Let us now take a moment to review how the more-conventional count reading is derived in a formal semantic system per the literature. Descriptively, there are two components to an FP’s semantics. First is a condition of parthood; the referent of a fraction partitive must be a proper part of the individual denoted by the complement.⁴ Secondly, there is a comparison between a degree of the referent and a degree of the complement on the same scale. The measure of the individual denoted by the FP on some scale is the measure of the complement on the same scale divided by the denominator. Intuitively, if a rock is three kilos, a third of that rock will be one kilo. Momentarily putting aside the issue of the measure reading, the main challenge in providing an account of the conventional behavior of FPs is their selectional flexibility. Consequently, the truth conditions of an FP are constrained non-trivially depending on the argument it takes.

Let us first examine the parthood component. Importantly, FPs can take either singular or plural individuals as their complement. This comes as no surprise as it can likewise be observed in many partitive constructions in many of the world’s languages (Moltmann, 1997; Wagiel, 2021). This can be seen transparently for Italian in (11), but it also extends to languages like Passamaquoddy (Algonquian), which is very much unlike the European languages we’ve been considering up to this point.

- (11) un terzo {della mel-a / delle mel-e}
a third {of.the.F.SG apple-F.SG / of.the.F.PL apple-F.PL}
‘A third of the apple/apples’

- (12) a. epahsiw cikon wekine
half apple.PROX.SG spoil.AI
‘half of the apple spoiled’
b. epahsiw cikoni-yik wekini-yik
half apple-PROX.PL spoil.AI-PROX.PL
‘half of the apples spoiled’

[Passamaquoddy]

Thus, a formal definition for FPs requires a shared type between singular and plural DPs. I follow Link (1983) et seq. in treating pluralities as individuals (type e) rather than as sets ($\langle e, t \rangle$), cf. Bennett 1974; van der Does 1991; Winter 2001). Borrowing heavily from Classical Extensional Mereology (Lesniewski, 1931; Leonard and Goodman, 1940), he treats pluralities as mereological sums of “atomic” individuals (singularities), which are of course also type e .

- (13) a. $[[\text{wall}]]^c = \{a, b, c, d\}$
b. $[[\text{the walls}]]^c = \bigoplus [[\text{wall}]]^c = a \oplus b \oplus c \oplus d$

This nets the selectional flexibility for FPs that we seek, but there is a lingering problem. In this system, membership of a plurality is modeled with the mereological relation (PROPER) PART-OF ($\sqsubset$).⁵ However, Link distinguishes two kinds of parthood which relate different kinds of objects. One, INDIVIDUAL *part-of*, holds between pluralities and the (sums of) individuals which compose them. The other, MATERIAL *part-of*, holds between singular individuals and their parts. This is done in service of the observation that predicates can apply to singularities but not necessarily the sum of their parts; as the famous example goes, a new ring can be made of old Egyptian gold. So, while both pluralities and singularities are type e , following Link’s system would predict that FPs—which presuppose parthood—could only be compatible with one or the other, else fractions are ambiguous between a singularity-selecting form and a plurality-selecting form.

⁴Some works (e.g., Ionin et al. 2006; Magri 2008) present an analysis of partitives s.t. *of* and its equivalents are the locus of the parthood assertion. While this is a plausible analysis, I do not follow this in the present work because—as far as I am able to tell—it confers no analytical advantage and defining fractions with parthood baked-in is notationally simpler.

⁵In a system that treats pluralities as sets, this would naturally be captured with a set-membership or (proper) subset relation.

This flies in the face of Moltmann and Wagiel’s observation about partitive constructions, prompting them to abandon the distinction between individual and material part-of.⁶ For the same reasons, I follow suit. Now, $\sqsubseteq$ is what holds between my material parts and me as well as me and any plurality of which I am a member. However, this makes a seemingly problematic prediction about the kinds of interpretations plural-containing partitives can have. I briefly touch on it here.

As readers familiar with CEM already know, even the simplest mereologies assume that parthood has the property of being transitive (Varzi, 2003) (14). If we maintain this for natural language, we would have to accept that the referent of a partitive containing a plurality could be a material part of an member of the plurality. For example, let’s say there is a plurality of New Yorkers of which Jerry Seinfeld is a member. Of course, Seinfeld’s right hand is a part of him, and he is part of the New Yorkers by virtue of being a member of that plurality. This would predict that Seinfeld’s right hand is in the extension of *part of the New Yorkers*, yet Seinfeld’s hand and nothing else being completely covered in paint is clearly not a verifying condition for (15). Let us call this the TRANSITIVITY PROBLEM.

(14) *Transitivity*

$$\forall x, y, z. (x \sqsubseteq y \wedge y \sqsubseteq z) \rightarrow x \sqsubseteq z$$

(Part of part of a whole is itself part of that whole.)

(15) Part of the New Yorkers are completely covered in paint

Moltmann and Wagiel seek to restrain the meaning of plural-containing partitives to pick out only singularities and their sums, excluding the material parts. To avoid this overgeneration, they posit that the transitivity of parthood can fail under certain circumstances. Specifically, for any given pair of individuals, one can only be part of another if there is no intervening individual which is an INTEGRATED WHOLE (16). This is what Moltmann and Wagiel call the TRANSITIVITY PRINCIPLE and TOPOLOGY-SENSITIVE TRANSITIVITY, respectively.⁷

(16) $\forall x, z. x \sqsubseteq z \rightarrow \neg \exists y. x \sqsubseteq y \wedge y \sqsubseteq z \wedge \text{int-wh}(y)$

How an integrated whole is defined varies between the two authors. Moltmann appeals to R-INTEGRATED WHOLES (Simons, 1987), which are defined in (17). To Wagiel, an IW is an individual that has the mereotopological property of being MAXIMALLY-STRONG SELF-CONNECTED (Casati and Varzi, 1999) relative to some property. Essentially, an MSSC individual is one who is SELF-CONNECTED (any two ways of dividing an individual are connected), has an interior which is self-connected, and the self-connected things which overlap with it are part of it. A formal definition is given in (20). Regardless, the end result is the same—a materially-real, single, whole individual like Jerry Seinfeld constitutes an IW whose parts are not members of any plurality said IW is part of.

(17) An entity x is an R-integrated whole if there is a division of x such that every member of that division stands in the relation R to every other member and no member bears R to anything other than members of the division. (Moltmann, 1997, 24)

(18) Self-connected

$$\text{SC}(x) \stackrel{\text{def}}{=} \forall y \forall z [\forall w (\text{overlap}(w, x) \leftrightarrow (\text{overlap}(w, y) \wedge \text{overlap}(w, z))) \rightarrow \text{connected}(y, z)]$$

(x is self-connected iff any two parts that form the whole of x are connected to each other.)

(19) Strongly self-connected

$$\text{SSC}(x) \stackrel{\text{def}}{=} \text{SC}(x) \wedge \text{SC}(\text{interior}(x))$$

(x is strongly self-connected if x is self-connected and the interior of x is self-connected.)

(20) Maximally-strongly self-connected

$$\text{MSSC}(x) \stackrel{\text{def}}{=} \text{SSC}(x) \wedge \forall y [\text{SSC}(y) \wedge \text{overlap}(y, x) \rightarrow y \sqsubseteq x]$$

With parthood out of the way, let us consider how to model the comparative component of fractions. To begin, I introduce to our ontology DEGREES (type d), which we may conceptualize as points on scales of measurement. Many natural language phrases arguably denote degrees, like heights (2 meters), monetary values

⁶For additional work which assumes a single parthood relation for individual and material parts, see Magri (2008); Sudo (2025).

⁷Moltmann’s view on what a part can be goes beyond the transitivity principle. She also notes that in certain cases it can depend on the whole itself what the parts are, which is explored in Moltmann (1998, prep).

(\$38), temperatures (21°C), and even cardinalities (*twelve*). It is standardly assumed that degrees can participate in arithmetical relations like *greater than*, *less than*, *equal to*, etc, and that at least some arithmetical operations (addition, subtraction, multiplication) may apply to degrees as well. I further assume, following Coppock (2022), that degrees may also participate in division (cf. Bale and Schwarz 2022, 2026). Degrees are retrieved from individuals via MEASURE FUNCTIONS (MFs, $\langle e, d \rangle$). There are a number of natural language items which plausibly involve measure functions in their truth conditions, most prominently gradable adjectives and measure words. Sample denotations for such items are given in (21), (22).

$$(21) \quad [[\text{tall}]] = \lambda d_d \lambda x_e. \mu_{\text{height}}(x) \geq d$$

$$(22) \quad [[\text{weigh}]] = \lambda d_e \lambda x_e. \mu_{\text{weight}}(x) \geq d$$

In these cases, the scale on which the argument of the measure function is evaluated is lexically-specified; *weigh* returns a degree on the weight scale, for example. However, a semantic analysis of fraction partitives demands something less specified. Forgetting momentarily about partitives with plural complements, the measure function of a fraction must be somewhat flexible; compare *a third of the apple*, where the relevant dimension of comparison could be volume or mass, and *a third of the movie*, where the dimension is likely runtime. Even the same argument can be measured in distinct ways depending on the situation; compare the two uses of *half of the apple* in (23).

- (23) a. Paint half of the apple $\rightsquigarrow$ surface area
b. Use half of the apple for the pie $\rightsquigarrow$ volume, mass

In this sense, fraction partitives resemble *more*-comparatives, which have been subject to considerable inquiry. As can be seen in (24), *more* is used in a comparison of degrees from two different scales. To capture this fact, it and other quantity words (like *much*) have been analyzed as introducing a VARIABLE MEASURE FUNCTION (VMF, represented μ^8) whose scale of measurement is not specified when it enters the derivation (Bale and Barner, 2009; Wellwood, 2019). Wagiel ultimately adopts μ for his account of the semantics of FPs.

- (24) a. Newman spun more yarn than Jerry $\rightsquigarrow$ length
b. Kramer listened to more music than George $\rightsquigarrow$ runtime

How VMFs are defined is not a free-for-all, though. Two important constraints have been identified to produce a successful analysis of, e.g., quantity words. For starters, the resulting MF must be MONOTONIC as defined in (25) (Schwarzschild, 2002, 2006; Nakanishi, 2004, 2007; Wellwood, 2019). In prose, a monotonic MF is sensitive to mereological relationships—a proper part of an individual measures less on a monotonic scale than the whole. By this definition, volume, weight, length, runtime, and cardinality are all monotonic scales. On the other hand, temperature, density, concentration (of, e.g., alcohol or salt in a solution), and speed are non-monotonic scales. Such scales ought to be excluded for defining VMFs given the kinds of interpretations comparative constructions like (26) can have.⁹

- (25) S-monotonicity Adapted from Wellwood (2019)
A measure function μ_m is S-monotonic if $\forall v, v'. v \sqsubseteq v' \rightarrow \mu_m(v) \leq \mu_m(v')$

- (26) Jerry drank more wine than Elaine
a. ✓ Jerry drank two glasses while Elaine drank one
b. #Jerry's wine was 45°F while Elaine's was 42°F

Let us now turn to the second constraint. Bale, Barner, and Wellwood propose that if the argument of a VMF is a plurality, the resulting MF must be a cardinality function. This position is motivated by a series of observations about how pluralities are interpreted in comparative constructions across a variety of languages and morpho-syntactic contexts. Experimental studies by Barner and Snedeker (2005) and more recently Le et al. (2025) have found that collections of discrete objects are consistently evaluated by number (rather than size or weight) in

⁸...without a subscript.

⁹This constraint extends beyond defining VMFs. As Krifka (1989) notes, heads of measure pseudopartitives can only be monotonic (also called EXTENSIVE) measure functions.

(i) {five ounces/*twenty carats} of gold

quantity comparison tasks. This is true even of object-mass nouns like *silverware* or *furniture*, which are like more prototypical mass nouns in refusing plural morphology and numerals. These results extend to languages like Japanese and Mandarin, where (morphological) mass-count and singular-plural distinctions are not as clear-cut as they are in English (Inagaki and Barner, 2009; Cheung et al., 2010).;

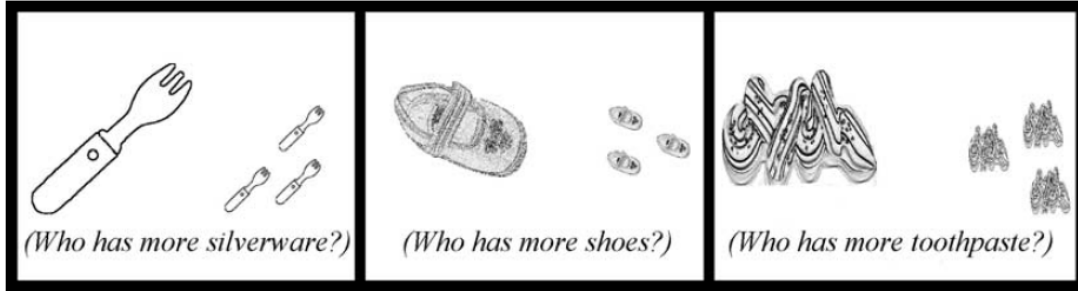


Figure 3: Experimental stimuli from Barner and Snedeker 2005, p. 50

This can be observed for pluralities of events as well as entities. (Wellwood, 2019, ch. 6) notes that certain “high-attachment” comparative constructions evoke a reading that is sensitive to the number of occasions rather than degree on some other scale. This data might encourage us to bias VMFs towards cardinality functions over other forms of measurement.

- (27) Ann was happy more than Bill was.
✓ more occasions, #more happiness

Bale & Barner implement these constraints as follows. They posit a partially-ordered set of measure functions which is composed of every monotonic measure function. Furthermore, cardinality is outranked by every other form of measurement in this set. This set, which I call M , is shown in (28).

$$(28) \quad M = \{\mu_{card} < \mu_{weight}, \mu_{volume}, \mu_{mass}, \mu_{length}, \mu_{runtime}, \dots\}$$

Bale & Barner further posit an algorithm for determining the scale of a VMF from the set of functions M . Defined in (29), a VMF is defined as the minimal MF in M which ranges over the argument it has taken. If there is no single minimal MF, context decides which MF it will be defined as. Since a cardinality function necessarily ranges over pluralities by making reference to atoms/integrated wholes, this algorithm guarantees a plurality will be counted when it is fed to a VMF.

- (29) Contextual conditioning
 μ is interpreted as one of the measure functions m_z in the series $\langle m_1, m_2, m_3, \dots, m_n \rangle$ such that the argument for μ is in the range of m_z ; furthermore, contextually m_z is preferred to m_y if $m_z < m_y$

Let me now briefly demonstrate how this system works. Imagine a VMF has taken an atomic individual/integrated whole like an apple as its argument, as in (30). This individual has many properties we expect a physical object to have—it has a weight, height, width, surface area, density, volume, and so forth. Any of the measure functions which correspond to these concepts could be applied to the individual apple and return a defined, nonzero degree. However, there are measure functions which do not range over the individual. It obviously does not have a runtime or degree of happiness, for example. For our purposes, we will further stipulate that a cardinality function cannot range over it.¹⁰ We now take the set of MFs which are defined for the apple and intersect it with M (28), the result shown in (31).

¹⁰It is too strong to say that a singularity like the apple does not have a cardinality. If Kramer only has a single apple, an inquiry into the degree of apples Kramer owns will obviously return the value *one*. In the original formulation of this algorithm, VMFs combine with sets $\langle e, t \rangle$ rather than individuals (e); cardinality functions in particular are only defined for join semi-lattices which include atomic individuals (Bale and Barner, 2009, §5.3.1). This rightfully excludes bare NPs (which are sets of atomic individuals but not their sums) and mass nouns (which are atomless semi-lattices, being divisible *ad infinitum*.) Wagiel (2021)’s adaptation treats MFs as taking a type- e argument, which is necessary if we believe fractions are combining with DPs but obfuscates the straightforward singularity-plurality distinction. We can patch this by additionally stipulating that the identity of a VMF must range over the arguments of every MF in the FP; cardinality is defined for a single apple but not a part of it, whereas cardinality is defined for a plurality of apples as well as the IWs and arbitrary sums of IWs which compose it.

$$(30) \quad \mu([[\text{the apple}]]^c)$$

$$(31) \quad \{m : m \in M \wedge m([[\text{the apple}]]^c) \neq \#\} = \{\mu_{card} < \mu_{weight}, \mu_{volume}, \mu_{mass}, \mu_{length}, \mu_{runtime}, \dots\}$$

The resulting set contains multiple MFs, none of which are minimal with respect to one another. Thus, context will ultimately pick the identity of the VMF from the set of remaining candidates. Once it has been defined, the function will return a degree along the relevant scale.

How do things play out when the argument of the VMF is a plurality? Consider such a case with a plurality of apples, for example. Once again, we intersect M with the set of MFs which are defined for the plurality. Importantly, the plurality is still defined for non-cardinal MFs; this is irrefutable given how lexically-specified measure functions can combine with pluralities and have cumulative interpretations (32). Yet, they also have a cardinality by virtue of being a plurality. The resulting set (33) now has a single minimal member, the cardinality function. Per the algorithm, the VMF will necessarily be a cardinality function. Context no longer has a say in the matter.

$$(32) \quad \text{The chairs (altogether) weigh 200 kilograms.}$$

$$(33) \quad \{m : m \in M \wedge m([[\text{the apples}]]^c) \neq \#\} = \{\mu_{card} < \mu_{weight}, \mu_{volume}, \mu_{mass}, \mu_{length}, \mu_{runtime}, \dots\}$$

With this machinery in place, we define fractions like *third* in (34). It takes an individual y and returns a set of individual x s with two constraints imposed. Firstly, x must be a proper part of y . Secondly, the degree of x must equal the degree of y multiplied by the fraction (in this case, $\frac{1}{3}$). The identity of the relevant degree is subject to the algorithm I sketched out above. To combine this set with predicates, it may first be taken by a choice function or t -operator (producing type e) or combining with an existential quantifier ($\langle\langle e, t \rangle, t \rangle$). The latter of these is most likely what is observed in more prototypical uses of fractions, given what appears to be an indefinite article (*a, un, ...*) at the left-edge.^{11 12}

$$(34) \quad [[\text{third}]] = \lambda y_e \lambda x_e. x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3}$$

To sum up this section, we designed and motivated a semantics for the more-conventional count reading of FPs which is flexible enough to deal with both singularities and pluralities. Tools from mereology, plus a constraint on parts of pluralities (i.e., sums of integrated wholes) characterize the parthood component. Contextually-defined measure functions which are additionally biased towards counting for pluralities handles the degree comparison component. Specifically, the count reading follows because of the ordering of the set in (28). Of course, these are insufficient for capturing the measure reading we observe in Italian, etc. In the following section, I propose an amendment to the machinery laid out in this section which ultimately produces the measure reading for Italian FPs.

3 Proposal

Briefly, I reduce the measure/count contrast to the presence or absence of a special operator which I call a MEASUREMENT SHIFTER (M-SHIFT). This operator, which I reason is covert in Italian, takes the set denoted by the fraction plus the complement and returns a new set, making it type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$. M-SHIFT changes how variable measure functions are evaluated to produce the desired truth conditions for measure FPs, effectively changing which parts of the complement get to be denoted by the fraction. Assuming the Y-model of grammar

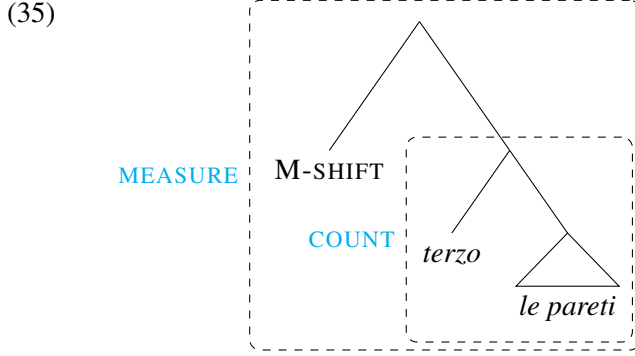
¹¹At least in English, introducing a universal quantifier to a sentence as in (i) also containing an FP permits both co-varying and non-co-varying readings. This lends additional credence to the quantifier option.

(i) Every member of the audience punched a third of the comedians.

¹²How might we derive fractions with numerators greater than one? One option would be to treat the fraction as any other countable NP. First, we apply Link (1983)'s $*$ -operator to the set, yielding a join semi-lattice of lone thirds, arbitrary sums of those thirds, and the maximal sum of every third. The set could then combine with a numeral (*two*), restricting the set to only entities which are the sum of exactly two "third"-entities. The second option would be to abstract over the numerator and have a numerator combine directly with the fraction. The latter is especially attractive since there seem to be at least some uses of fractions where the numerator and denominator form a constituent.

(i) The company hired [two-thirds] New Yorkers

where syntax builds structure before shipping it to interpretational and phonological components, I assume that the operator combines with the FP at Narrow Syntax. I reason that this produces the agreement effects that we see. The structures are schematized in (35).



Let me spell out transparently what is happening on the interpretation side. Remember, Bale and Barner’s algorithm guarantees that pluralities will be counted rather than measured via their ordering of the set M by which VMFs are defined. I change this state of affairs by treating M as an intensional parameter by which logical forms are interpreted (as opposed to some ambient element of cognition/grammar). Doing so makes M shiftable, like other well-defined intensional parameters (worlds, times, contexts, and so forth). The guilty party in this case is M-SHIFT, which switches the “matrix” or default setting M to M' . M and M' have the same members, but the ordering relation is reversed—in this latter, cardinality is the maximal measure function.

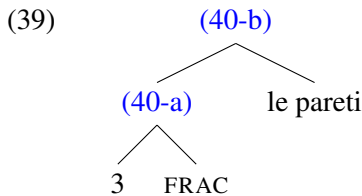
I define M-SHIFT as in (36), taking a predicate of individuals and returning another where the set of evaluation is M' (37) rather than the “matrix” set M . As a result, measure functions in the scope of M-SHIFT will be defined with the Bale & Barner algorithm relative to M' instead of M . While M and M' are identical in terms of their members, the ordering relation has been reversed so that cardinality is now maximal with respect to other forms of measure. Per the algorithm, variable measure functions will be non-cardinal even if they are ranging over a plurality.

$$(36) \quad [[\text{M-SHIFT}]] = \lambda P_{(e,t)} \lambda x_e. [[P(x)]]^{M'}$$

$$(37) \quad M' = \{\mu_{\text{weight}}, \mu_{\text{volume}}, \mu_{\text{mass}}, \mu_{\text{length}}, \mu_{\text{runtime}}, \dots, < \mu_{\text{card}}\}$$

Let us now walk through the derivation step by step. I assume a semantics for fractions partitives generally as shown in (38). This function takes a cardinal number n and returns a two-place predicate which is true if x is part of y and the degree of x is the degree of y divided by n .¹³ I follow Wagiel in assuming that the preposition/genitive case is semantically vacuous. By default, the measurement parameter is set to M . The less-marked count reading proceeds as in (39)-(40), presumably with an additional step of universal quantification or application of a choice function to allow the structure to combine with predicates.

$$(38) \quad [[\text{FRAC}]]^{c,M} = \lambda n_n \lambda y_e \lambda x_e. x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{n}$$



$$(40) \quad \begin{array}{ll} \text{a.} & [[\text{terzo}]]^{c,M} = \lambda y \lambda x. x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3} \\ \text{b.} & [[\text{terzo delle pareti}]]^{c,M} = \lambda x. x \sqsubset \oplus [[\text{wall}]]^c \wedge |x| = |\oplus [[\text{wall}]]^c| \times \frac{1}{3} \end{array}$$

Once the fraction combines with a plurality, the ordering of M dictates that the VMF will be a cardinality function. Returning to the six walls of the studio apartment in §1, the conditions of (40-b) determine that the

¹³In Italian and many other languages, denominators (outside of suppletive forms like *half*) are derived from or exactly homophonous with the number’s corresponding ordinal. Given this generalization, it is fair to wonder whether the meaning of a fraction is derived from the meaning of an ordinal. Such an account for FPs has been given before (López Palma, 2011, 2012) but I abstract away from this detail since it has little bearing on the semantic alternation we are interested in.

members of the extension will be the two-wall sums which are part of the wall plurality. The cardinality function is insensitive to individual variation in wall size, making any two-wall sum a possible referent of the count FP. This is schematized in *fig. 4*.

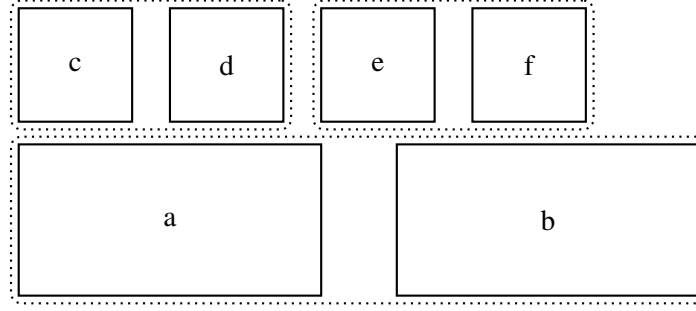
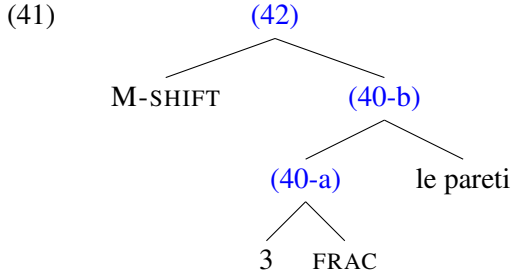


Figure 4: Partial extension of count FP (walls not to scale).

To get the measure reading, M-SHIFT applies to this structure and changes M to M' (41)-(42). By the same algorithm, VMFs in the scope of the operator will never be cardinality functions since it is now the maximal member of the set of MFs. Now, even a plurality can be measured along a non-cardinal dimension.



$$(42) \quad [[\text{M-SHIFT terzo delle pareti}]]^{c, M'} = \lambda x. x \sqsubseteq \bigoplus [[\text{wall}]]^c \wedge \mu(x) = \mu(\bigoplus [[\text{wall}]]^c) \times \frac{1}{3}$$

Let us imagine that in this case, context has determined that the relevant MF is a function of surface area. Given a plurality of walls whose cumulative surface area is $12h$ square meters, the resulting third must be $4h$ square meters. Ergo, the referent could be one of the large walls (a or b) or the sum of the small walls ($a \oplus b \oplus c \oplus d$), as in *fig. 5*.¹⁴

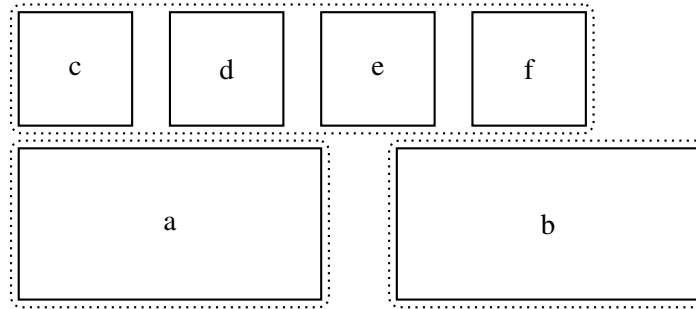


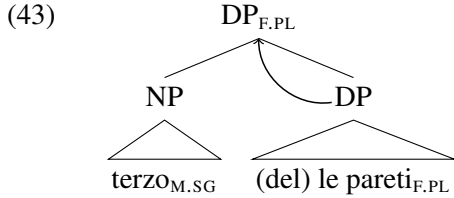
Figure 5: Partial extension of measure FP (walls not to scale).

How does this analysis derive the agreement alternation which introduced this puzzle? I would like to suggest that the observed facts can be boiled down to an intervention effect. As stated previously, M-SHIFT is an operator which is introduced in Narrow Syntax rather than post-syntactically at LF. This is not inherently problematic as many other proposed intensional operators transparently enter at NS. *Believe* and other doxastic predicates are exemplary cases, shifting the world of evaluation from the actual world to a possible world for

¹⁴For this proposal to work for *all* cases of the measure reading (see §1, p. 2), transitivity suspension as outlined in §2 must ultimately be abandoned. Now, even a part of a single member of a plurality can be part of a plurality. However, this creates a seemingly problematic prediction about what, e.g., *part of the boats* can refer to. While I omit the details here for sake of brevity and relevance, there is some reason to believe that the problematic readings of plural partitives are excluded on pragmatic grounds rather than semantic. I direct curious readers to [Roberts \(2025\)](#) for further details.

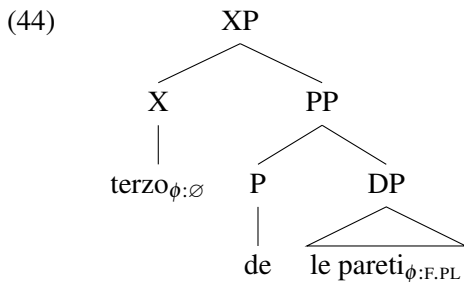
(parts of) the embedded clause (von Fintel and Heim, 2011). Having M-SHIFT in the syntax means it can feed or bleed strictly-syntactic operations, like Agree.

First, we must discuss how to capture agreement with count FPs. In this case, which I am calling the less-marked of the two, agreement is with the complement of the FP. There are two imaginable ways to model this syntactically. Assuming a model where features on a head project to the maximal projection, it could be that the complement is actually heading the FP. This has been suggested for a number of partitive constructions as early as Selkirk (1977), and it give the correct characterization of the agreement facts without refuting the “featurefulness” of fractions; in Italian and beyond, fractions seem to take number and gender suffixes. Having the complement project respects this fact while getting the correct predictions.



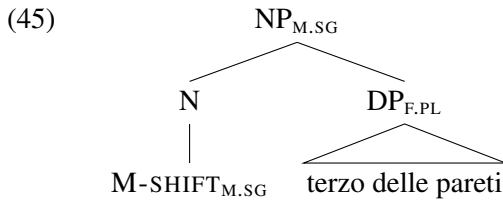
However, there are a number of problems with this approach. Positing such a structure for FPs is unusual given that the complement appears to be introduced to this construction via the preposition *di*. Ergo, to say the complement DP projects its label would necessitate that the preposition projects no syntactic structure. This is not necessarily the death knell as it could be explained as a case-marker in the style of Chomsky (1970), but this begs the question of where this *di* case comes from. If genitive/partitive case is typically assigned by a nominal to its complement, the fraction is the clearest candidate for assigning case. Yet, the present circumstances preclude case assignment by the fraction because the FP complement is the thing which projects (ergo, the fraction does not actually take a syntactic complement).

An alternative would be to say that the fraction is blind to agreement processes in some way, in either bearing no features at all or lacking a feature a probe on T can interact with (Deal, 2015). With this configuration, a probe will continue to look down into the FP structure, eventually finding and copying the features on the complement. This derives complement agreement while maintaining a more-familiar structure for partitives. Klockmann (2018) suggests such an analysis for English quantity pseudo-partitives which appear to host number morphology such as {*a lot/lots*} of *New Yorkers*. In these cases, agreement is always with the internal nominal rather than the quantity word.



This nets the desired result but at the expense of a straightforward explanation for fraction morphology. Furthermore, we do not presently have good reason to believe that this number morphology is not the number/gender morphology we see on typical nominals. As Klockmann demonstrates, there is strong evidence for *-s* on *lots/tons of NP* not being a genuine plural suffix. For one, the distinction between *a lot of* and *lots of* is semantically arbitrary; we cannot confidently conclude this for fractions since the number suffix is tightly controlled by the numerator of the fraction (**un terzi*, **due terzo*). We also observe the indefinite article *un* showing concord with the fraction {**una terzo*}, a fact that is hard to explain if fractions are truly featureless.

I remain agnostic to the internal structure of count FPs. What is crucial is that they are configured s.t. the features of the complement are copied. I suggest that M-SHIFT is on some level a nominal which comes with a MASC feature and projects its label upon combining with the count FP structure. This state of affairs makes the masculine feature more local to a probe than whatever feature the count FP bears, so complement agreement is impossible. As a result, what I have been calling fraction agreement is actually agreement with M-SHIFT.



Treating the operator as a covert, feature-bearing noun introduces a nominal syntactic category to the left-edge of the FP which, on either account of complement agreement I sketched above, the count FP structure lacks. This ends up being a fortunate result; in Italian, certain nominal modifiers on the fraction preclude sentential agreement with the complement. This is exemplified in (46), where the relative clause restricts the fraction.

- (46) Il [terz-o del-le paret-i dipint-o da John] {è ross-o / *sono ross-e}
 the.M.SG [third-M.SG of-the wall-F.PL painted-M.SG by John] {be.3SG red-M.SG / be.3PL red-F.PL}
 ‘The third of the walls painted by John are red’

To summarize our discussion up to this point, I resolve the count-measure alternation to a covert piece of functional structure which (i) is nominal, bearing a gender feature, and (ii) shifts a parameter *M* to manipulate how VMFs are defined. It is the presence of this functional item that transforms a count FP into a measure FP. An astute reader may wonder, though, whether the cline in markedness I posit is justified. For starters, it is not clear that one FP is semantically more marked than the other. To illustrate, let us consider a transparent case. Markedness at the level of interpretation can be thought of as strength of meaning, which is itself a function of how few possible worlds it is true in. For example, *Newman is a mailman* is weaker than *Newman is a sinister mailman*. This is because the former is true in all worlds where Newman is a mailman, whereas the latter is true only in worlds where Newman is a mailman *and* sinister (which is of course a subset of the former set). Correspondence between semantic markedness (stronger meaning) and morphosyntactic markedness (more words/morphemes) is the *modus operandi* many assume, something that has come to be known as HORN’S DIVISION OF PRAGMATIC LABOR (Horn, 2001; de Swart and Farkas, 2010).¹⁵

As I illustrated in §1, though, the two readings for Italian FPs are disjoint from one another—neither is a subset of the other. Granted, there can be substantial overlap between the measure and count reading of an FP in contexts where the individuals which compose the relevant plurality are congruent; in these cases, a subplurality can be a third by mass or by cardinality. However, the disunity of the two readings is made clear when we control for this. Regardless, the circumstances are such that no reading is stronger than the other. Additionally, FPs in Italian are superficially identical within the DP domain regardless of the reading being count or measure. On both counts, we have no support for the markedness cline I posit.

I maintain my position in spite of this because it confers analytical benefits for typology. To begin with, treating the measure FP as the sum of the count FP and an operator yields a straightforward explanation for some of the crosslinguistic variation observed in §1. As I discuss there, American English lacks the measure reading altogether. Relatedly, agreement is necessarily with the complement if an FP contains a plural DP. If the measure reading and its corresponding agreement pattern resolve to M-SHIFT (this proposal), American English can be defined as a language which simply lacks the operator. Thus, the difference between this language and Italian, etc. is a simple matter of lexical variation.¹⁶

Secondly, there are languages where the measure reading can be overtly more marked than the count reading. It is seen in (8) that Russian exhibits an agreement-interpretation similar to that in Italian; however, this language additionally exhibits a FP construction which includes the part-word *čast*’. Notice in (47) that only the measure interpretation is available and agreement is necessarily with the fraction/*čast*’. In light of these facts, we might interpret *čast*’ as an overt realization of M-SHIFT.¹⁷

¹⁵...but this is not universally adopted. Perhaps most famously, plurals have been argued to be semantically-less marked than their singular (bare) counterparts in English and other languages (Sauerland et al., 2005).

¹⁶This approach has the additional benefit of accounting for gaps in FP interpretation. There are languages where plural-containing FPs have a measure and count interpretation (Italian), and there are languages where such FPs only have a count interpretation. To my knowledge, however, there are no languages where a language has access to the measure interpretation to the exclusion of the count interpretation. Deriving the measure reading with an operator which is itself parasitic to the count reading makes these languages impossible, since the existence of the measure reading entails the count reading.

¹⁷This point requires some qualifications. Despite appearances (and how it is glossed), I do not take this use of *čast*’ to mean *part* in the more-familiar sense. Some languages have superficially similar FPs (i.e., ones which include an overt *part*-word) yet they exhibit completely different behavior. English has (or at one point had) a construction similar to Russian *čast*’ FPs, exemplified in (i). It

- (47) tret'ja čast' sten krasn-{\check{a}ja/*ye}
 third.ADJ.F.SG part.F.SG wall.F.PL red-{F.SG/PL}
 'A third part of the walls are red'
 ✓M, ×C

[Russian]

Thus concludes my proposal for a treatment of the agreement-interpretation alternation for Italian FPs. The application of an operator M-SHIFT to the structure shifts a parameter of interpretation that would otherwise force pluralities to be measured along a cardinality scale. I further suppose that this operator creates an environment where agreement with the complement is impossible, deriving “fraction” agreement. My choice of treating the measure reading as more marked than the count reading allows for a clean explanation of American English FPs as compared to Italian, etc. In the following section, I discuss what other pathways for a theory of the alternation and compare them with the proposal given in this section.

4 Comparing alternatives

Here I present three possible alternatives to my own proposal. The first is not truly an account of FPs, but presents a way of implementing agreement/interpretation correlation that is interesting enough to warrant discussion. The second are proposals from works about partitives more generally, but they offer treatments for deriving what I have been calling the measure interpretation. I highlight the merits and demerits of each and compare them with my own proposal.

4.1 Alternative implementation: semantic agreement?

As sketched in §3, the agreement facts for Italian FPs follow from probes copying the features of a goal given some structural configuration; i.e., they are products of the Agree operation. The presence of M-SHIFT, I argue, introduces a new bundle of features which gets copied over the features of the complement. However, there is a conceivable alternative explanation for the agreement facts which I discuss here. This alternative offers some analytical advantages over my syntactically-present operator analysis, at least *prima facie*. Ultimately, though, the approach is found to be untenable.

In a recent proposal, Charnavel and Sportiche (2025a,b) argue that the features on agreeing heads (e.g., T/Infl) are at least sometimes semantically-contentful rather than uninterpretable copies of features (contra the original formulation of Agree in Minimalism, where features copied by a probe are necessarily not interpreted at LF). They entertain this possibility to account for what are often called *semantic agreement* phenomena, characterized by an apparent mismatch between the features on a nominal and an agreeing verb. Crucially, the verb's exponed features affect how the subject is interpreted—e.g., whether *old* holds of the members or the whole in (48), whether the speaker is amongst the majority in (49). In the syntax, Charnavel and Sportiche argue that the feature on T introduces an additional presupposition about the subject (it is a plurality, it is first person, etc.) and the meaning of the overall sentence is calculated with this contribution. I dub this approach INTERPRETABLE INFLECTION.

- (48) a. The committee is old [British English]
 ~> the organization was formed a long time ago ∨ the members of the committee are old
 b. The committee are old
 ~> the members of the committee are old, ↯ the organization was formed a long time ago
- (49) 'The majority of us are always here' [French]

It is difficult to comprehensively evaluate the semantic import of *part* in these FPs because they seem quite marginal in contemporary Englishes; as of 2025, the Corpus of Contemporary American English (CoCA, Davies 2010) contains 7613 instances of *a third of* but only 23 instances of *a third part of*. Similarly, the British National Corpus (BNC, Davies 2004) contains 949 instances of *a third of* but a mere 3 instances of *a third part of*. Semantics aside, (i) shows that at minimum the presence of *part* does not preclude agreement with a complement, unlike in Russian.

- (i) [A] third part of the waters **were** turned into wormwood (CoCA, Davies 2010)

The details of English FP-*part* constructions are interesting and warrant further investigation. Chief among these questions is what the semantic difference between this and bare FPs might be, if any. However, it is beyond the scope of the present work. The agreement facts are enough to dissuade us from an analysis where the measure reading arises from the addition of *part*-words.

- a. *une majorité d'entre nous {est/sont} toujours là*
a majority among 1PL COP.{3SG/3PL} always here
↪ speaker may or may not be part of the subject
- b. *une majorité d'entre nous sommes toujours là*
a majority among 1PL COP.1PL always here
↪ subject must be part of the subject

At first glance, the alternation presented in §1 and the kinds of cases represented by (48)-(49) appear remarkably similar; both exhibit an interpretive alternation conditioned by morphology on an external element. I must emphasize here that Charnavel and Sportiche do not consider the FP data in their work, and (to my knowledge) they do not seek an explanation of the topical alternation. Regardless, though, the striking similarities encourage an analysis of Italian FPs that is an extension of interpretable inflection. Pursuing this route has the potential to yield analytical fruit. Most obviously, we would gain a straightforward explanation for why a difference in verbal inflection affects how FPs are interpreted; in my proposal, the agreement facts are the result of an operator intervening between a probe and would-be goal, an arguably indirect explanation. Additionally, interpretable inflection could let us abandon the notion that the measure reading and count reading correspond to distinct syntactic structures. As it stands now for Italian, the measure reading is derived by application of an operator that we cannot see. We need not entertain this possibility under a Charnavel and Sportiche-type account; measure FPs and count FPs look the same because they *are* the same. The locus of interpretive differences is outside of the FP.

While successful for prototypical cases of semantic agreement, a closer examination of the topical alternation reveals that this approach is untenable. For starters, both measure and count readings are available for fraction partitives outside of positions of agreement—this was seen for Italian in (2), repeated below. Under interpretable inflection, it is the features on the agreeing head which licenses additional interpretations. This account predicts that at least one of the two readings would be inaccessible outside of subject position, yet this is not realized.

- (50) *la muffa ha coperto un terz-o del-le paret-i*
DET.F.SG mold has.3SG covered.3SG one third-M.SG of-DET.F.PL wall-F.PL
‘The mold has covered a third of the walls’
✓M, ✓C

Additionally, I noted in §3 that certain modifiers of the fraction (e.g., relative clauses) in the FP preclude agreement with the FP’s complement (51). There, I took it as auxiliary evidence that M-SHIFT introduces nominal characteristics that make “fraction” agreement possible. More generally, though, it supports the notion that measure FPs are structurally distinct from count FPs, despite appearances. While this is a natural consequence of my proposal, interpretable inflection predicts that either agreement option should be possible regardless any modifiers inside the FP. This is because the locus of the alternation is on T/Infl, which (according to standard assumptions of syntax) should say nothing about the kinds of modifiers the subject can receive.

- (51) *Il [terz-o del-le paret-i dipint-o da John] {è ross-o / *sono ross-e}*
the.M.SG [third-M.SG of-the wall-F.PL painted-M.SG by John] {be.3SG red-M.SG / be.3PL red-F.PL}
‘The third of the walls painted by John are red’

Finally, interpretable inflection is successful with regards to grammatical number only if the alternation is conditioned by singular and plural number on the verb. In an account which pursues interpretable inflection, a singular feature on T/Infl somehow introduces a presupposition which yields the measure reading and a plural feature on T/Infl does the same for the count reading. Empirically-speaking, this is not an accurate characterization of the current data. The measure interpretation is licensed by agreement with the features on the fraction, not necessarily singular. Conversely, the count interpretation is licensed by agreement with the features of the complement, not necessarily plural. If this were not the case and the alternation was indeed sensitive to simply singular versus plural, it would predict that predicates would permit singular agreement with FPs containing larger fractions, like two thirds. As seen in (52), though, singular agreement with *due terzi delle pareti* ‘two thirds of the walls’ is significantly-marked if not ungrammatical.

- (52) ??*due terz-i del-le paret-i è copert-o di muffa*
two third-M.PL of-the wall-F.PL be.3SG covered-M.SG by mold

Int. ‘Two-thirds of the wall are covered by mold’

Despite its initial appeal, the above data dissuades us from pursuing a treatment of the fraction partitive alternation as semantic agreement. My analysis, in contrast, is one where the agreement facts are due a distinct syntax between the two, with semantic distinctness following from the tenet of ONE FORM/ONE MEANING.

4.2 Previous suggestions for deriving measure FPs

While the count reading for FPs has received the majority of the attention of the literature, the measure reading has not gone completely unnoticed. Something like it has been suggested at least since Hoeksema (1996) who argues that a FP like *half of Jane and Jacky* most naturally refers to the sum of half of Jane and half of Jacky rather than one of them individually.¹⁸ However, the fact that in some languages the measure reading is licensed only by a certain agreement configuration is an original observation; as far as I am aware at least, no author up to this point has made the connection. Here, I explore two proposals and evaluate how they fare for the set of data on which this work focuses.

4.2.1 Type ambiguity for plural DPs

In her account of the Partitive Constraint, de Hoop (1997) proposes that (most) plural DPs are ambiguous between individuals (e) and sets ($\langle e, t \rangle$). She further posits that some partitives differ in what kind of semantic type they select for; most relevant to us, the heads of proportional partitives (fractions, but also percentages, etc.) can take either type as an argument. Thus, the difference in meaning is resolvable to the semantic type of the complement. Under her account, there are no restrictions on what part of a plurality can be (cf. the discussion of transitivity suspension in §2), whereas subsets must be comprised of individuals in the relevant set. In de Hoop (2003), she writes:

I assume that Jane and Jacky can denote a complex entity, which explains the grammaticality of *half of Jane and Jacky*[.] At the same time, I conclude that *Jane and Jacky* cannot denote a set of entities and this accounts for the ungrammaticality of **one of Jane and Jacky*[.] Note that *half of Jane and Jacky* does not refer to Jane or Jacky. If we consider *Jane and Jacky* to denote a composed entity, then half of it can be any half.

Given what she has said up to this point, a fraction must be able to take sets and individuals. To handle this, we could plausibly posit higher types (something which is neutral between e and $\langle e, t \rangle$) and posit a relation to stand in for both *part-of* and *subset-of* (Lewis, 1991). She does not say anything about how the relevant measure function would be defined, but I conjecture there could be a rule that makes cardinality the only viable measure for a set argument.

De Hoop’s account gets us both readings—the count reading arises when the complement is interpreted as a set, whereas the measure reading arises when the complement is interpreted as an individual. However, a number of issues can be identified. In light of the cross-linguistic data discussed in this work, de Hoop’s analysis does not lend itself to a clean account of the typology. If American English does not have a measure reading, de Hoop would have to say that the relevant plural DPs can only denote sets in this language; can de Hoop commit to this position? Falco and Zamparelli (2019) (discussed further in the following subsection) present a series of arguments against her position. For one, treating definite plurals as $\langle e, t \rangle$ opens them up to semantic processes typically reserved for NPs and predicates (quantification, predicate modification, etc.), making forms like **red the flowers* possible without further stipulation. Furthermore, what has the privilege of $e \sim \langle e, t \rangle$ ambiguity in her system is ill-defined. She argues that coordinated singulars can only be e , yet a coordinated DP like *the North*

¹⁸You will notice here and in the remainder of the section that these explorations of the interpretation of FPs cite English data. I personally disagree with many of their judgments for my own English and maintain that the measure reading for FPs is not robustly attested in American English. For cases involving conjunction, the preferred reading is very plausibly a case of conjunction reduction (*half of Jane and half of Jacky*) and thus distinct from the measure reading we’ve discussed so far. Falco and Zamparelli (2019) (discussed at length in §6.2) report similar readings for FPs containing definite plural DPs, but as far as I am able to tell neither of them are native American English speakers. I have shown that the measure reading is observed in many of the languages spoken in Europe including Italian, so I suspect judgments from their respective L1s are influencing how they can interpret these FPs. I discuss the works regardless because their descriptions of the phenomena match that of measure readings and their proposed solutions are plausible approaches to analyzing measure FPs.

Pole and the South Pole picks out the same plurality as *the Earth's poles*, a definite plural—which can be either e or $\langle e, t \rangle$ according to de Hoop. All of these points cast doubt on the type ambiguity position.

4.2.2 (Covert) count-mass shifting

In explaining the measure reading for FPs, [Falco and Zamparelli \(2019\)](#) end up taking an entirely different path for deriving the measure reading. Coincidentally, their suggestion pairs well with the method for deriving count FPs sketched in §2. Recall that the reason the measure reading was ungenerable under this system was the bias to counting in defining VMFs; μ will be a cardinality function if it takes a plurality as its argument, per the algorithm proposed by Bale & Barner. If we want to hold this constraint constant, one play might be to “trick” the fraction into thinking its complement isn’t really a plural individual.

Falco & Zamparelli take exactly this approach. Specifically, they suggest that the measure reading is the result of a covert operator which takes the plural complement and returns unindividuated mass—perhaps the UNIVERSAL GRINDER of [Pelletier \(1975\)](#); [Pelletier and Schubert \(1989\)](#). They write (p. 36):

The only remaining ingredient to obtain mass proportions is the presence of a count-to-mass type-shift (from an object to its material constituent part), which seems to be independently needed in light of many mass uses of count nouns, well documented in the literature[.]

We can define a UG as in (53), which takes an individual x and returns a new individual which is the generalized sum of the set of z s s.t. z is a part of y and y is a proper part of x .

$$(53) \quad [[UG]] = \lambda x. \bigoplus \{z : \forall y. z \sqsubseteq y \wedge y \sqsubset x\}$$

If we assume transitivity suspension (§2), this operator will return an object which is distinct from the original individual. For starters, while Jerry Seinfeld’s hand is not a part of the New Yorkers, his hand is a part of $[[UG]]([the\ New\ Yorkers])$. Furthermore, if the means of individuating atoms/integrated wholes has been destroyed, it is now impossible to count the members of a plurality. Thus, to Falco & Zamparelli, the structure of the measure FP is that in (54).

$$(54) \quad [\text{terzo} [\text{UG} [_{DP} \text{le pareti}]]]$$

The appeal of this proposal is that, at first blush, it requires virtually no additional assumptions. It allows us to maintain the algorithm for defining variable measure functions, which seems robust given Bale & Barner and Wellwood’s observations as discussed in §2. Furthermore, the exceptional reading—the measure interpretation, which prompts my revision to VMF-defining—is derived via an off-the-shelf mechanism which may be independently needed for count-to-mass coercion processes. This is all done without positing type ambiguities for plural DPs as de Hoop does.

However, there are a number of flaws with this suggestion despite the initial allure. I take this moment to discuss each in turn. To begin with, many languages with a grammatical number distinction on nominals allow mass nouns to be “bare”—i.e., without non-singular number morphology or articles. The same is true for count nouns which have been coerced into having a mass interpretation; in French and English, for example, canonically-count nouns appear bare under a mass interpretation (55)-(56), and the same holds for Italian (57). This is straightforwardly explainable if we assume count-to-mass operations apply to NPs (sets of atoms/IWs) instead of DPs (58) ([Rothstein 2009](#); a.o.). Such an operation, I conjecture, would block the application of at least plural morphology.

$$(55) \quad \begin{array}{l} \text{Il} \quad y \quad a \quad \text{assez de tapis} \quad \text{pour nous quatre} \\ \text{EXPL there have.3SG enough of carpet for} \quad \text{1PL four} \\ \text{‘There is enough rug for the four of us’} \end{array} \quad [\text{French}] \text{ (Adapted from } \a href="#">Bosque 2024, p. 139)$$

$$(56) \quad \begin{array}{l} \text{The three squirrels carelessly wandered into traffic and were struck by a motorist.} \\ \text{Now there is squirrel all over the road.} \end{array}$$

$$(57) \quad \begin{array}{l} c' \text{-}\grave{e} \quad \text{banan-a} \quad \text{sul} \quad \text{paviment-o} \\ \text{EXPL-be.3SG banana-F.SG on.the.M.SG floor-M.SG} \\ \text{‘There’s banana on the floor’} \end{array}$$

$$(58) \quad [[UG]] = \lambda P_{\langle e, t \rangle} \lambda x_e. \exists y. y \in \pi_i(P) \wedge x \sqsubset y \quad \text{Adapted from } \a href="#">Kiss et al. (2021)$$

However, under a measure interpretation of an FP—which F&Z would analyze as the result of a UG applying to the complement—we observe no change on the morphology of the complement in Italian. It is not impossible that the operator we would need to capture the measure reading is distinct from that which is responsible for more-familiar count-mass conversions. This would predict, though, that mass interpretations of plural DPs could be found elsewhere in Italian if context made required it. This is because there is nothing stopping $UG(DP_{PL})$ from occurring outside of a fraction partitive. At present, I am aware of no such evidence.

More damningly, though, measure FPs in Italian are still compatible with predicates which necessarily make reference to atoms. If the measure reading is truly resolvable to a covert shifting of the plural complement to a mass, we expect this kind of reference to be impossible; measure FPs should only be compatible with predicates which are insensitive to atomic structure. This would follow from Falco & Zamparelli’s position on how the measure reading is derived—the members of the plurality cannot be counted because the shift from count to mass makes counting impossible. Ergo, they would say VMFs are not defined as cardinality functions. This is of course the case with canonical mass nouns and in prototypical cases of count-to-mass conversion, shown below.

- (59) a. *two water
b. #The water is parallel/elliptical/stacked.
- (60) Context: The army of robots were smashed to pieces by the anti-AI protesters.
a. *They swept two robot into a bin.
b. #Robot is (each) begging for mercy.

By contrast, Italian measure FPs may combine freely with lexically distributive predicates. One such predicate is *ellittico* ‘elliptical’, seen in (61). In this case, we observe the predicate agreeing with the features of the fraction; this is of course a known characteristic of measure FPs. A verifying case for (61) is shown in *fig. 6*; there are four ellipses, one square, and one triangle. However, each ellipses is much smaller than the square or triangle; their cumulative area is only a third of the the cumulative area of all the shapes; therefore, they will be selected for by the measure reading of *un terzo* but not the count reading, since $\frac{4}{6} \neq \frac{1}{3}$. Despite the fact that this is clearly the measure reading, the predicate *ellittico* can still be true of it. Since *ellittico* necessarily holds of IWs, measure FPs must be able to have IWs as part of their interpretation.

- (61) un terz-o delle figur-e è ellittic-o
a third-M.SG of.the.F.PL shape-F.PL be.3SG elliptical-M.SG
‘A third of the shapes are elliptical’

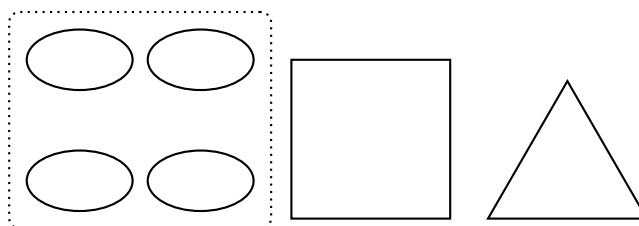


Figure 6: Four ellipses in a collection of six shapes (not to scale).

The same point can be made with predicates containing distance distributivity operators (i.e., derived distributive predicates). In (62), the Italian distributivity marker *ciascuno* ‘each’ forces the predicate to hold of every IW which is part of the subject.¹⁹ As a consequence, that subject must be composed of IWs (i.e., they are not masses). Measure FPs (with their respective agreement pattern) may still be subjects such predicates, further proving that they are indeed pluralities. A verifying case for (62) can be found in *fig. 7*, where four out of six cakes are topped with a strawberry, but said four have a cumulative volume worth one third of the cumulative volume of all the cakes together.

- (62) un terz-o delle tort-e è decorat-o con una fragola ciascun-a
a third-M.SG of.the.F.PL cake-F.PL be.3SG decorated-M.SG with a.F strawberry each-F.SG
‘A third of the cakes were decorated with a strawberry each’

¹⁹(62) is complicated by the fact that the distributive operator has feminine agreement. I am not certain why this is, but crucially, the verb agrees with the fraction.

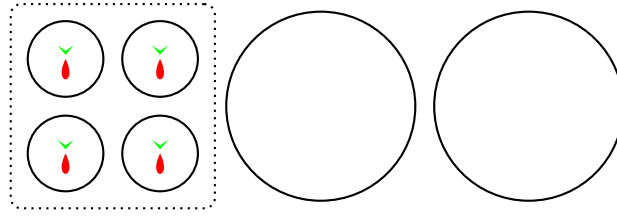


Figure 7: A collection of six cakes, four of which are topped with a strawberry (not to scale).

One final illustrative example comes from discrete collective predicates like *accatastato* ‘stacked’.²⁰ Such predicates are like those modified by ‘each’ in necessarily selecting for a plurality while making reference to integrated wholes in said plurality; ergo, they cannot be combined with masses. Yet, they may take fraction partitives as their subject while copying the features of the fraction, as shown in (63). This sentence can be true in contexts like *fig. 8*, where four cakes out of six are arranged in a stack, yet these four cakes together have a volume which is one third of the total volume of cakes.

- (63) un terz-o delle tort-e è accatastat-o
 a third-M.SG of.the.F.PL cake-F.PL be.3SG stacked-M.SG
 ‘A third of the cakes are stacked’

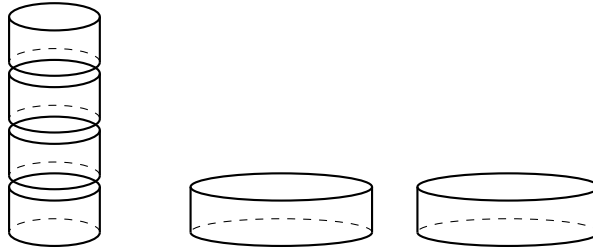


Figure 8: A collection of six cakes, four of which are stacked on top of each other (not to scale).

Were this data not enough, I direct the reader to (64)-(65), two examples taken from the Italian-language corpus PAISÀ (Lyding et al., 2014). In these cases we see FP subjects of predicates which range over (animate) integrated wholes; (65) is particularly interesting because it contains a possessive adjective anaphoric to the individual members of the plurality. In both cases, we see the agreement pattern characteristic of measure FPs. F&Z’s suggestion would predict them to be impossible with these kinds of predicates, but once again this is contrary to fact.²¹

- (64) un terz-o delle guardi-e aveva un comportament-o ritenut-o sadic-o
 a third-M.SG of.the.F.PL guard-F.PL had.3SG a behavior-M.SG consider-M.SG sadistic-M.SG
 ‘One-third of the guards exhibited behavior considered sadistic’

²⁰ Astute readers may worry if this particular predicate is genuinely a discrete collective predicate or if it is a mass/plurality-selecting predicate analogous to English *piled (up)*. Observe that this predicate cannot combine with genuine masses like *neve* ‘snow’.

- (i) #la neve è accatastat-a
 the.F.SG snow be.3SG stacked-F.SG
 Int. ‘A third of the snow is piled up’

²¹ With regards to (64)-(65), it is difficult to imagine that the original author meant to pick out a subset of the guards/British women on the basis of anything besides cardinality. For example, if there are 300 guards, the author likely means to say that 100 guards exhibited arguably sadistic behavior. It seems unlikely that a subset of the guards whose cumulative volume or mass is a third of that of all the guards is the subset exhibiting sadistic behavior. For one, delimiting the guards this way prerequisites knowledge of their individual volumes/masses. The original context of these sentences and their authors’ intent will likely remain a mystery by virtue of this being corpus data, but the “100 out of 300” reading is reasonably the more likely of the two.

I maintain the position that these examples are genuine measure FPs. Recall in §1, fn. 1 I mentioned that in contexts where the members of a plural complement are congruent (same size and shape), there is overlap in the denotation of measure and count FPs. What I conjecture is happening in (64)-(65) is the authors have assumed a context where individual variation along certain degree scales is negligible. Thus, the very same plurality which is a third of the guards by measure could also be a third of the guards by count.

- (65) ne emerge che un quint-o delle donn-e inglesi vorrebbe un “design”
 EXPL comes.out C a fifth-M.SG of.the.F.PL woman-F.PL English.PL would.like.3SG a design
 nuov-o per [il propri-o naso]
 new-M.SG for the.M.SG own-M.SG nose
 ‘It turns out that one-fifth of British women would like a new “design” for their nose’²²

To summarize, measure FPs in both novel and corpus data have been found to be compatible with predicates which range over (sums of) integrated wholes. I take this as strong evidence that measure FPs can denote pluralities. They can also denote things besides pluralities of course (see “third of each”-cases discussed in §1), but this ability is left unexplained by Falco & Zamparelli’s suggested proposal. If the measure reading is resolvable to something like a universal grinder as they say, its application should bleed combination with a distributive predicate. This is because F&Z’s suggestion depends upon the destruction of the plural DP’s “individuation structure” in order to block the count reading from applying. Distributive predicates and the like rely on this individuation structure to range over their arguments, thus predicting a ban on the cooccurrence of measure FPs and distributive predicates. As I have shown in (61)-(65), these predicted downstream effects are not realized, leaving us with a proposal which overgenerates.

Allow me to conclude my commentary on Falco & Zamparelli’s proposal with a remark about these kinds of coercion processes and cross-linguistic variation. Count-mass coercion is indeed quite robust in English, the language whose data initially prompted the proposal of the universal grinder in Pelletier (1975). Its ubiquity puts the *universal* in *universal grinder*. Beyond the bounds of English, though, this alleged mechanism is not so universal. Perhaps most famously, Cheng et al. (2008) show that IW/atom-denoting nouns in Mandarin (e.g., *gǒu* ‘dog’) do not permit mass-ish interpretations. Note that Mandarin is an articleless, numberless language, so we expect no morphological changes like those we see in English count-mass coercion.

- (66) qiáng-shang dōu shì gǒu
 wall-TOP all COP dog
 ✓ ‘There are dogs all over the wall’
 × ‘There is dog all over the wall’ [Mandarin]

Italian is a language with some nouns which, despite canonically being count, can also be coerced to a mass noun. This was seen for Italian *banana* in (57). However, fieldwork up to this point suggests that this is a particular property of certain nouns rather than a general property of Italian syntax-semantics. This is illustrated in (67), where two canonical count nouns like *scoiattolo* ‘squirrel’ and *calamaro* ‘squid’ do not permit mass readings.

- (67) #c’-è {scoiattol-o / calamar-o} sul paviment-o
 EXPL-be.3SG squirrel-M.SG / squid-M.SG on.the.M.SG floor-M.SG
 Int. ‘There’s squirrel/squid(-mass) on the floor’

While I stop short of a comprehensive investigation of Italian count nouns and their compatibility with measure FPs, measure FPs seem to be quite productive. We have seen them take pluralities countable, physical objects like *torte* and *pareti*, but also more abstract pluralities like *figure* and *film*. This casts an additional layer of doubt on F&Z’s proposal, since (i) the data in (67) might suggest a UG-type mechanism does not exist in Italian, and (ii) if the few mass-count coercions Italian does permit are indeed the work of such a mechanism, this should delimit the possibilities for measure FPs. This second point seems contrary to fact.

4.3 Summary and advantages of present proposal

In §3, I present an analysis which yielded both measure and count FPs. I assume that count FPs are unmarked case and derive the measure reading via an operator M-SHIFT which makes cardinality functions inaccessible when defining a measure function. The operator is assumed to be syntactically present and bearing its own features, thus deriving “fraction” agreement under the measure reading. In this section, I present perspectives which represent fundamentally-different paths an alternative analysis could follow.

²²Because the original was suggestive, I changed the last noun and the gender of the determiner, possessive adjective for concord reasons.

These perspectives have their own advantages which make them enticing; as I hope I have made clear, though, many of them ultimately suffer from problems of over/undergeneration or difficulty in parameterization. Interpretable inflection, while appealing given the topical alternation's superficial similarity to semantic agreement phenomena, falls flat since it offers no explanation as to why both readings are accessible when there is no agreement. In contrast, both readings are generable under my system because it follows from the application of an operator within the DP domain; it can show up regardless of argument position. It just so happens to feed a particular agreement pattern when it is in a position of agreement, e.g. [spec,TP].

We also saw two specific suggestions for explaining measure FPs from de Hoop and Falco & Zamparelli, who are couching their analysis in broader conversations about the semantics of partitives. De Hoop argues that the ambiguity is a result of ambiguity for DPs between sets and entities, but she offers little evidence for this position outside of the domain of partitives. My analysis, in contrast, does not place the onus of interpretational change on the complement; it is an operation specifically on the FP which derives the measure reading. As a result, I make no predictions about ambiguous readings of plural DPs outside of FPs. This is a welcome result, since to my knowledge there are none.

Finally there is the proposal from F&Z, which coincidentally dovetailed well with standard assumptions about how the count reading is derived. They suggest we resolve this ambiguity to a covert count-mass conversion mechanism. As I discuss at length, though, this method ends up under-generating for Italian. If measure FPs are actually just FPs which have taken a “massified” complement, they should be barred from combining with distributive or discrete collective predicates. As it turns out, there is no problem with such configurations of FPs and predicates in Italian. This is all made worse for the current set of data when we observe that count-mass coercion is not a very productive process for nominals in Italian. On the other hand, the current account permits measure FPs with distributive predicates because the fundamental structure of the complement remains the same. The only thing which distinguishes the count reading from the measure reading is how VMFs are defined; thus, a measure FP may still refer to a plurality.

5 Concluding discussion

To conclude, this work identifies an agreement-interpretation alternation found in several Indo-European languages and provides an account for Italian specifically. A novel contribution to the description of these items, it is reported that these readings can be disambiguated with certain agreement configurations; agreeing with the features of the FP's complement mandates a count reading, while copying the features of the fraction permits at least a measure reading. I ultimately present an account starring a special operator M-SHIFT which changes how measure functions are defined at LF. This proposal is developed to also explain typological facts (why American English lacks the measure reading but no known language represents the reverse case). Alternatives such as semantic agreement, semantic type ambiguity, or the application of a semantic type-shifter are considered, but I present a series of arguments for each showing that they do not provide a satisfying account.

Outside of the empirical contribution and its corresponding analysis, the primary theoretical contribution of this work lies in measurement. Of the two readings, measure is deeply mysterious, given the discussion in §2. As I write there, a variable measure function which takes a plurality as its argument is traditionally defined as a cardinality function. This position has been a long-standing one of degree semantics; Bale & Barner et seq.'s finding that comparative constructions which range over pluralities robustly make evaluations by cardinality is a convincing reason. On the face of it, the mere fact that the measure reading exists seems to spell trouble for this tenet. In light of this, Falco & Zamparelli's suggestion for deriving the measure reading is particularly attractive. If they are right, we may maintain standard assumptions about how VMFs; to them, the complements of measure FPs are not genuine pluralities, so they cannot be counted. As I discussed at length, though, there are reasons to think that measure FPs do refer to pluralities.

I ultimately derive the facts by (i) treating the set of possible measure functions as a parameter which can be shifted, and (ii) introducing intensional operators in Narrow Syntax to shift them. This is admittedly a great deviation from established thought on degree semantics; however, I defend it as the best analytical option. In the face of FP data in Italian, etc., one could take the nihilistic position that there are no constraints on VMF-definition besides monotonicity—everything is purely contextual, regardless of the mereo-topological status of the argument. This is obviously wrong given the observed robustness of cardinality in comparative constructions, both across semantic type and across languages. This is also wrong in the microcosm of Italian FPs; the measure reading does not arise arbitrarily but is constrained by a particular configuration in positions of agreement.

As I see it, my proposal is a compromise between the status quo that Bale & Barner establish (too strong, undergenerates) and what we might call METROLOGICAL NIHILISM (too weak, overgenerates). Measuring a plurality via a VMF is possible, but it is contingent on M-SHIFT—which is not present in all of the world’s languages, if my theory of cross-linguistic variation is correct. Otherwise, the “matrix” ordering for measure functions is used.

As far as I know, there are no prior suggestions of this type in the literature, and skepticism is warranted. While the measurement-shifting story outperforms, e.g., Falco & Zamparelli’s analysis, it is fair to wonder if this is the best answer given its degree of specialization and, conversely, its lack of generalizability to other phenomena. For this analysis to hold water, it seems necessary to identify other phenomena which (i) are resolvable to VMFs, (ii) mandate other forms of measurement over counting, and (iii) are necessarily products of elements introduced in Narrow Syntax (as opposed to, e.g., the context). Whether or not these items exist is the object of future research.

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